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PALEOMAGNETISM AND ROCK MAGNETISM OF THE MISSISSIPPIAN LEADVILLE (CARBONATE) FORMATION AND IMPLICATIONS FOR THE AGE OF SUB-REGIONAL DOLOMITIZATION

Robert A. Horton, Jr. and John W. Geissman

Department of Geology, Colorado School of Mines

Richard J. Tschauder

Hecla Mining Company

Abstract. The Mississippian Leadville Formation of Central Colorado is dolomitized near the Sawatch Uplift and hosts numerous Pb-Zn-Ag mineral deposits. Paleomagnetic and rock magnetic data from several dolomite types away from centers of Tertiary intrusive activity imply a late Paleozoic (Late Mississippian, ?) age for the dolomitization (e.g., D = 148°, I = -11°, k = 86, α_{95} = 8.6°, N = 3 localities, 31 samples in dolomitized units). The magnetization is of diagenetic origin and was probably acquired during or soon after dolomitization. Tertiary thermochemical activity has overprinted dolomites close to intrusive centers. Data indicate that limestones removed from igneous activity were also remagnetized at the same time.

Introduction and Geologic Setting

The nature and origin of the paleomagnetism in carbonate rocks have received considerable attention over the past decade [e.g., Lowrie and Alvarez, 1977; Heller and Channell, 1979; Lowrie and Heller, 1982; McCabe et al., 1983a]. Paleomagnetic data have constrained the ages of events affecting carbonates, such as regional dolomitization, base metal mineralization, and oxidation [e.g., Wu and Beales, 1981; Shanks et al., 1979; McCabe et al., 1983a; Wisniowiecki et al., 1983; Channell et al., 1982]. It has been shown that the major carriers of the magnetization may be diagenetic in origin [McCabe et al., 1983a,b]. We have examined aspects of the Leadville Formation (Kinderhookean-Osagean) of Central Colorado. The Leadville Formation offers an opportunity to study the relations between diagenesis, regional dolomitization, Tertiary igneous activity, structural deformation, and the magnetization of a stable shelf carbonate. Here, we discuss the paleomagnetism of Early Mississippian limestones and sub-regionally dolomitized equivalents on the flanks of the Sawatch uplift (Figure 1). The spatial association among pervasively dolomitized Leadville Formation, numerous Pb-Zn-Ag deposits, and Laramide (latest Cretaceous to mid-Tertiary) plutons in the Leadville-Gilman area have led to suggestions that the intrusives were the source of the metals and that hydrothermal fluids were the source of magnesium in the dolomites [e.g., Thompson, 1982; Lovering et al., 1978]. Recently diagenetic and hydrothermal dolomites have been recognized [Tschauder and Landis, 1983]. The overall character of the ore deposits, which are found throughout the Leadville Formation, have prompted suggestions that some metals were introduced by low-temperature brines in Late Paleozoic time [DeVoto, 1983; Tschauder and Landis, 1983; Horton et al., 1983]. Understanding the age and origin of dolomitization is important to understanding the Pb-Zn-Ag deposits. This investigation is concerned with paleomagnetic information bearing on the age of dolomitization.

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Sampling and Laboratory Procedure

We report data from 10 locations around the northern Sawatch uplift (Figure 1). At 6 locations, more than 7 samples were collected over narrow stratigraphic intervals. In four cases, samples were collected through 30 m or so of a measured stratigraphic section.
As relief is considerable in the region, fresh, unweathered material was readily available.
All samples were collected by orienting blocks (independent measurements) in the field using a magnetic compass; standard specimens were prepared in the laboratory. We employed a spinner magnetometer, on which was attached a 500 count/rotation optical shaft encoder. Signal analysis and data acquisition were carried out using an interfaced microprocessor. The instrument has a sensitivity of better than 1×10^{-4} A/m; this is adequate to resolve almost all significant components during demagnetization. Orthogonal demagnetization diagrams and vector subtraction techniques were used to calculate component directions. Extraction of magnetic material for Curie temperature analysis and SEM inspection followed the methods of McCabe et al. [1983a] and Wisniowiecki et al. [1983].

Results

The Leadville units exhibit three types of behavior in demagnetization that relate to their geographic location and geologic history. Limestones near Minturn (Ml) and Lime Creek (LC) (Figure 1) have NRM intensities usually less than 5×10^{-3} A/m. On the site level, components are well-grouped (α_{95}'s are 8° and 7° for Ml and LC, respectively) and are of southerly declination and moderate negative inclination (Figure 2), although both polarities occur at the sites. IRM data imply that the dominant magnetic phase in these units is magnetite. Most dolomites in and near the Leadville-Gilman area also possess a Laramide affinity magnetization (Figure 2). This component often comprises over 90% of the NRM, but in some cases, it is superimposed upon components of

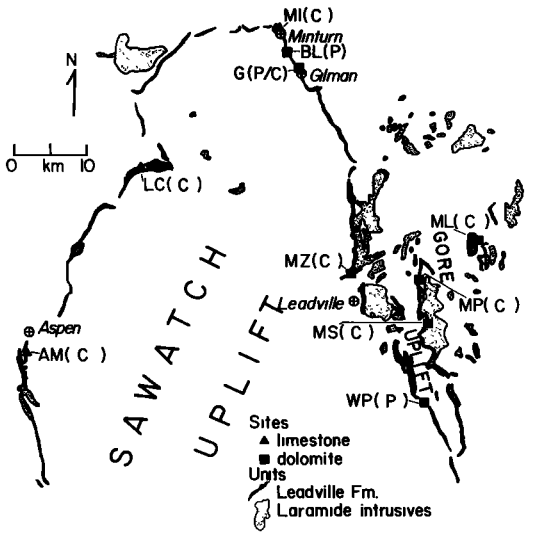


Fig. 1. Location map of the Leadville Formation, Sawatch uplift, and Laramide intrusives. The estimated age of the magnetization common to each sampling locality (P = Late Paleozoic, C = Cretaceous/Cenozoic) is given beside the sampling localities, which are as follows: AM, Aspen Mountain; LC, Lime Creek; Ml, Minturn; BL, Bolts Lake; G, Gilman; MZ, Mount Zion; ML, Mount Lincoln; MP, Mosquito Pass; MS, Mount Sherman; WP, Weston Pass.

higher coercivity and blocking temperature of southeasterly declination and shallow negative inclination. As with the limestones, both polarities of the Laramide affinity magnetization are common. NRM intensities often exceed 5 x 10^{-1} A/m in these units. IRM data again imply magnetite as the dominant magnetic phase, but hematite is also present. The third type of behavior, shown by many dolomite types (zebra, coarse, fine, breccias, etc.) is best displayed in the Weston Pass (WP) area and at Bolts Lake (BL) (Figure 1). The major component is of a shallow negative inclination and usually of a south-southeast to southeast declination, although some antipodal components have been recorded. NRM intensities are usually about 1 x 10^{-2} A/m for the units. Demagnetization response is favorable (Figure 3). Median destructive field values greater than 30 mT and IRM data imply single-domain and pseudo-single-domain magnetite as the dominant magnetization carrier. The age of the units and the directions of characteristic magnetization imply that the magnetization is of late Paleozoic age.

Discussion

Several lines of evidence suggest that the magnetization common to unheated dolomites was acquired during and soon after late Paleozoic sub-regional dolomitization. Magnetization properties of the unheated dolomites and limestones do not differ distinctly, although there is a slight tendency for higher median destructive field values and more discrete blocking temperatures in the dolomites. The directions of components in these lithologies are different (Figure 2). Also, magnetization acquisition spanned diagenetic or early epigenetic brecciation events. Samples from intensely brecciated zebra dolomite at both Weston Pass and Gilman yield well-grouped magnetizations identical to those of non-brecciated zebra dolomites (Figure 3), indicating that these dolomites must contain a secondary magnetization, rather than a depositional one. Support for this suggestion lies in the fact that closer to the center of igneous activity (Leadville), Laramide components reflecting thermochemical remagnetization are superimposed on late Paleozoic ones. Paleotemperature estimates of dolomites from the Weston Pass area (R.J. Tschauder and G.P. Landis, unpublished data, 1984) yield temperatures that are too low for a thermal remagnetization origin. Important evidence for a diagenetic origin of the remanence is also found in the magnetic extracts from dolomites and limestones. They contained spheroidal magnetite shapes which appear to be of diagenetic origin (Figure 4). Qualitative energy dispersive analysis on the SEM and thermomagnetic work indicate that iron is the only detectable cation present. As McCabe et al. [1983a,b] and Wisniowiecki et al. [1983] concluded, we suggest that the magnetite forms are not detrital in origin.

DeVoto [1983] and Tschauder and Landis [1983] proposed that subregional dolomitization of the Leadville Formation was an early diagenetic process related to Late Mississippian recurrent uplift prior to mineralization. Were magnetization acquisition to have occurred at this time, then in situ data should be corrected for post-Early

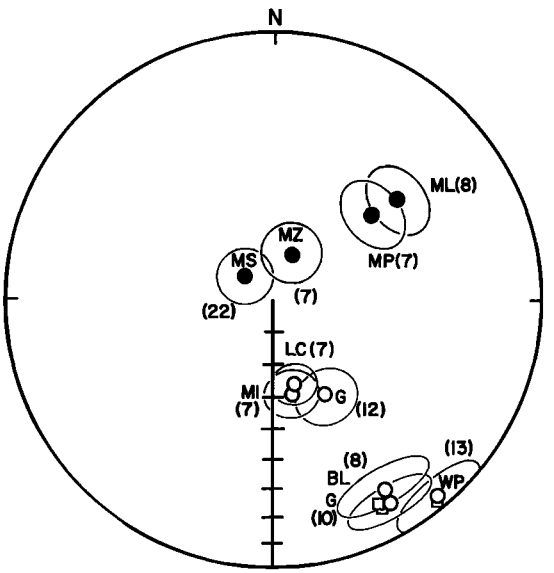


Fig. 2. Equal area stereographic projection of locality mean directions and projected cones of 95% confidence. Solid (open) symbols refer to lower (upper) hemisphere projections. The squares represent the late Paleozoic directions corrected for tilt. The data from Aspen Mountain (AM) were difficult to interpret and are not included. Identification of sampling localities follows Figure 1; the number of samples investigated, from which usually two specimens were demagnetized, is given in parentheses.

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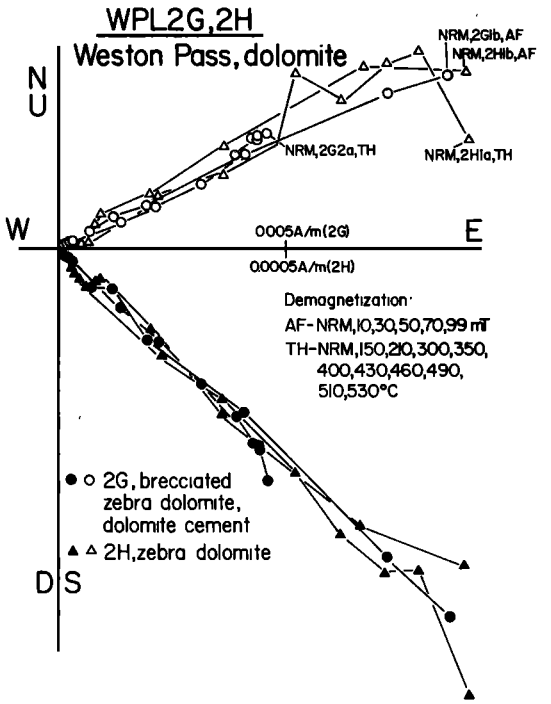


Fig. 3. Orthogonal demagnetization diagram [Zijderveld, 1967] emphasizing typical behavior of Leadville zebra dolomite and zebra dolomite breccia from Weston Pass. Open (closed) symbols refer to in situ projections on the vertical (horizontal) planes. Similar (Late Paleozoic) components are removed from both lithologies using either thermal or AF methods.

Mississippian deformation. Detailed work in the Sherman Mine has shown that the strata in some areas dipped to the east in Late Mississippian time (R.J. Tschauder, unpublished data, 1983).

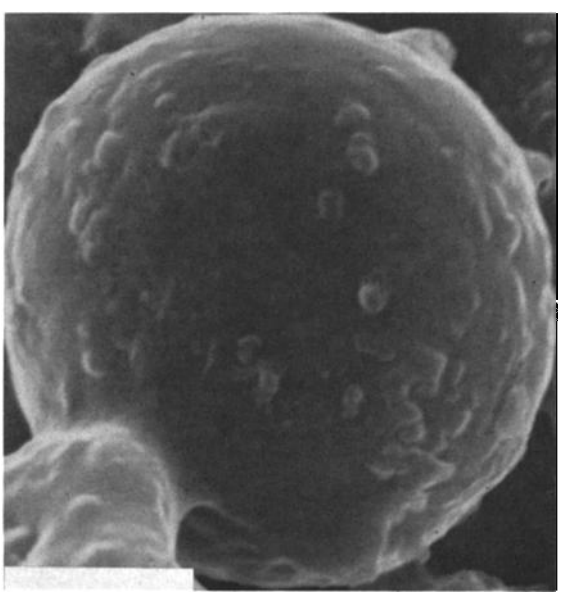


Fig. 4. Scanning electron microscope photomicrograph of magnetite sphere extracted from a a sample of dolomitized Leadville Formation. The scale bar is 2 microns.

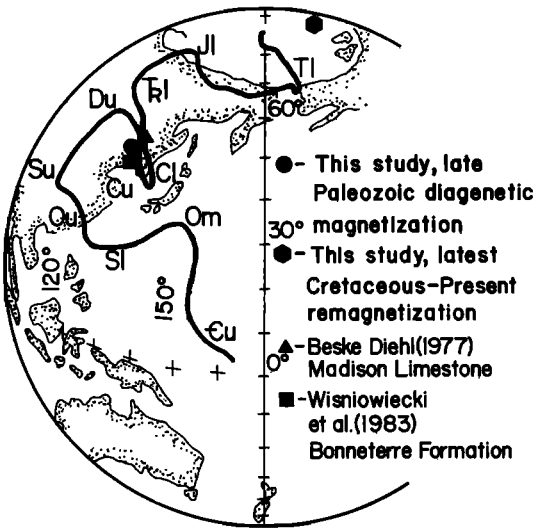


Fig. 5. Post Cambrian APW path for North America, showing paleomagnetic pole positions for the Leadville units and other Paleozoic carbonate formations discussed in the text. From Wisniowiecki et al. [1983] and Irving and Irving [1982].

Using corrections for minimum local tilt (about 11°), the paleomagnetic pole derived from the unheated dolomites lies along the Carboniferous segment of the North American apparent polar wander path (Figure 5). The data are as follows: D = 148°, I = -11°, α_{95} = 8.6°, k = 86, N = 31 samples from 3 locations; pole lat. = 46°N, pole long. = 123°E, dp = 4°, dm = 8°. For these units, this is the only span of post-Mississippian time during which the APW path has such a position and thus the magnetization must be late Paleozoic in age. The Early Carboniferous path, though not well-established, does not differ significantly from the Late Carboniferous segment. While the data are consistent with early diagenetic dolomitization, we cannot precisely date the event other than being late Paleozoic.

Dolomites closer to igneous activity reflect thermochemical resetting in Laramide time. Magnetite of diagenetic origin may still be preserved in these units, but thermal demagnetization behavior and high NRM intensities indicate that the magnetization is usually a TRM, or a partial TRM, rather than of chemical origin. Oxidation concomitant with igneous activity or paleoweathering often was significant; some of this secondary magnetization may be carried by hematite. The age of secondary magnetization, especially with respect to Laramide uplift and tilting, is equivocal. We are reluctant to apply corrections for regional eastward dip off the Laramide Sawatch uplift. Data from early Tertiary sills and stocks in the Gore-Mosquito Ranges east of Leadville (W.L. Oppenheimer and J.W. Geissman, unpublished data, 1984) show broad uplift to have been established prior to intrusive activity.

The preponderance of Laramide affinity magnetizations in the Leadville limestones provide an important distinction when compared to dolomites with late Paleozoic chemical remanent magnetizations. The magnetizations are probably not TRM's reflecting intrusives long since eroded away. For

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instance, the Pando Porphyry near Gilman is less than 4 m thick and likely never reached Minturn. In addition, no igneous rocks are present in the Lime Creek area, where limestones display well-grouped reverse polarity directions of Laramide affinity. We suggest that the magnetization carriers in the limestones were more susceptible to Tertiary/Recent effects. A very important factor in preserving diagenetic magnetizations in carbonate rocks may be the specific location of the secondary magnetic phases in the rock matrix. It may be that Leadville limestone at any location is incapable of acting as a high-fidelity recorder of an Early Mississippian field. In this context, our results are similar to those of Beske Diehl [1977]from the time-equivalent Madison Formation in Wyoming, where late Paleozoic magnetizations were best preserved in dolomitic units.

Conclusions

Magnetizations in unheated Leadville Formation dolomites are of diagenetic origin and were acquired in late Paleozoic time. Wisniowiecki et al. [1983] have proposed the regional significance of a late Paleozoic diagenetic magnetization event which affected Cambrian carbonates in southeast Missouri, prior to base metal mineralization. In the case of the Leadville Formation, we suggest magnetization acquisition was closely related to sub-regional dolomitization. Strong support has been given by several workers for spatial and temporal associations of karstification, dolomitization, and mineralization with late Paleozoic uplift within the ancestral Rocky Mountains. This idea contrasts with the theory of Tertiary dolomitization/mineralization. Our data cannot expressly serve to date the age of mineralization in the Leadville area, but the results do offer further compelling evidence for these associations.

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J.W. Geissman and R.A. Horton, Jr., Department of Geology, Colorado School of Mines, Golden, CO 80401.

R.J. Tschauder, Hecla Mining Company, Victor, CO 80860.

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